

Q/E - Greedy strategy for Multi-Armed Bandit

Given,

$$\epsilon = 0.2$$

$$N(A) = N(B) = N(C) = 0$$

$$Q(A) = Q(B) = Q(C) = 0$$

Formula:

$$Q_{\text{new}}(a) = Q_{\text{old}}(a) + \frac{R - Q_{\text{old}}(a)}{N(a)}$$

Play 1

Assume exploitation selects A

Rewards:

$$R = 3$$

$$N(A) = 1$$

$$Q(A) = 0 + \frac{3 - 0}{1} = 3$$

$$\therefore Q(A) = 3, Q(B) = 0, Q(C) = 0$$

Play 2

$$Q(A) = 3, Q(B) = 0, Q(C) = 0$$

Exploitation selects A again

$$R = 4$$

$$N(A) = 2$$

$$Q(A) = 3 + \frac{4 - 3}{2}$$

$$= 3 + \frac{1}{2}$$

$$Q(A) = 3.5$$

Play 3

A has the highest Q value.

$$Q(A) = 3.5$$

Next A reward:-

$$R = 2$$

$$N(A) = 3$$

$$Q(A) = 3.5 + \frac{2 - 3.5}{3}$$

$$= 3.5 + \frac{1.5}{3}$$

$$Q(A) = 3$$

Play I - A

$$Q(A) = 0 + \frac{3 - 0}{1}$$

$$= 3$$

Play 2 - A

$$Q(A) = 3 + \frac{4 - 3}{2}$$

$$= 3.5$$

Play 3 - B

$$Q(B) = 0 + \frac{6 - 0}{1} = 6$$

N Table

$$N(A) = 4$$

$$N(B) = 4$$

$$N(C) = 2$$

Reward Table

$$R(A) = [3, 4, 2, 5]$$

$$R(B) = [6, 7, 5, 8]$$

$$R(C) = [4, 3]$$

Q Table

$$Q(A) = 3.5$$

$$Q(B) = 6.5$$

$$Q(C) = 3.5$$

$$\therefore \text{Max}(3.5, 6.5, 3.5) = 6.5$$

Selected machine = B